

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 1						
DLT rate	0.3	0.35	0.4	0.45	0.5	
Efficacy rate	0.05	0.06	0.07	0.08	0.09	
Utility 2	-0.2	-0.3	-0.3	-0.4	-0.4	
Utility 3	31	29.6	28.2	26.8	25.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29.8	9.9	1.1	0	0	59.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.5	8.5	0.8	0	0	60.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	29.5	9.6	0.6	0	0	60.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.3	7.7	1	0	0	61
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30.1	8.2	1.6	0	0	60.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.1	9.1	0.9	0	0	60.9
U2 - Efficacy additive alpha prior Beta(1, 1)	30.4	8.2	1.4	0	0	60
U3 - Efficacy additive alpha prior Beta(1, 1)	29.9	8.9	0.9	0	0	60.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	5.7	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.8	5.4	1.4	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	5.7	1.4	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.9	5.4	1.4	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	5.7	1.5	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.9	5.3	1.4	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.5	5.7	1.5	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.9	5.4	1.4	0.6	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	3.1	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.1	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3	0.5	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.2	0.5	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1	3.1	0.5	0.3	0	
Scenario 2						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.2	0.25	0.5	0.5	0.5	
Utility 2	0.1	0.1	0.2	0.1	0.1	
Utility 3	46	47	58	56	52	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	29	24.8	36.2	8.5	0.8	0.7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	31.7	25.5	34.5	7	0.5	0.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.6	25.8	34.8	9.3	0.8	0.7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	32.1	26.5	33.4	6.8	0.5	0.7
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	29.1	25.5	34.7	9	1	0.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.9	24.9	32.7	7.2	0.5	0.8
U2 - Efficacy additive alpha prior Beta(1, 1)	29.7	25.4	33.8	9.4	1	0.7
U3 - Efficacy additive alpha prior Beta(1, 1)	33.1	25	33.4	7.3	0.4	0.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.4	8.5	8.2	2.5	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.9	8.7	7.7	2.3	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.5	8.4	8	2.6	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.3	8.6	7.6	2.2	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	8.3	8	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.3	8.3	7.8	2.2	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.5	8.2	8.1	2.7	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	11.4	8.3	7.7	2.2	0.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.7	1.6	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.8	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	1.6	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.6	1.6	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.6	1.7	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.6	1.6	0.7	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 3						
DLT rate	0.15	0.2	0.3	0.35	0.45	
Efficacy rate	0.1	0.4	0.4	0.4	0.4	
Utility 2	-0.1	0.2	0.1	0.1	-0.1	
Utility 3	40	56	52	50	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.8	51.6	25.3	7.5	0.8	3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.1	57.2	21.2	5	0.6	2.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.7	52.7	25.4	6.3	0.9	3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.7	57.5	21.1	5.4	0.3	3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12	53.7	23.3	7	1	3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.9	56.6	21.5	4.7	0.6	2.7
U2 - Efficacy additive alpha prior Beta(1, 1)	13	53	24.3	5.9	0.9	2.9
U3 - Efficacy additive alpha prior Beta(1, 1)	13.7	57.3	20.8	4.8	0.5	2.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8	12.4	6.4	2.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.2	13.2	5.8	2.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8	12.4	6.3	2.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.2	13	5.9	2.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	12.1	6.3	2.8	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.3	12.8	5.9	2.3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	8	12.1	6.4	2.7	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8.4	12.9	5.8	2.3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4.3	1.8	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.3	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	4.3	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.3	1.6	0.7	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	4.3	1.7	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.3	1.6	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	4.3	1.7	0.8	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.3	1.6	0.7	0.1	
Scenario 4						
DLT rate	0.05	0.1	0.18	0.3	0.4	
Efficacy rate	0.05	0.1	0.25	0.5	0.55	
Utility 2	0	0	0.1	0.2	0.1	
Utility 3	41	42	47.8	58	57	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.4	8.4	21.1	53.3	10.6	0.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	9.9	20.3	51.6	8.3	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	8.8	21.8	51.3	10.7	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	9.7	21.1	50.1	9.5	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.2	8.3	21.5	51.1	11.7	0.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	9.7	20.8	49.3	9.4	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	6.6	8.5	21.9	51.3	11.5	0.2
U3 - Efficacy additive alpha prior Beta(1, 1)	11.1	9.2	21.4	48.7	9.4	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	5.1	7.4	10.4	2.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.8	5.1	7.4	10.4	2.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.5	5.1	7.4	10.1	2.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.8	5.1	7.6	10.1	2.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	5	7.4	10	3.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5.1	7.6	10	2.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.5	5	7.5	10	3	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.9	5.1	7.5	10	2.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.7	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	1.7	2.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.7	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.1	1.7	2.5	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.1	1.7	2.5	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.2	1.7	2.5	0.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.1	1.7	2.5	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.2	1.7	2.5	0.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 5						
DLT rate	0.07	0.12	0.17	0.22	0.3	
Efficacy rate	0.05	0.3	0.3	0.3	0.3	
Utility 2	0	0.2	0.1	0.1	0	
Utility 3	40.2	53.2	51.2	49.2	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.5	35.9	26.5	23.3	9.4	0.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.4	43.1	25.2	17.9	6	0.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	36.4	25.4	24.5	8.6	0.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.2	43.2	24.6	18.7	5.9	0.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.3	34.6	27	24.4	8.3	0.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.8	43.4	23.6	19.3	5.5	0.4
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	35.2	26.5	24.3	8.5	0.4
U3 - Efficacy additive alpha prior Beta(1, 1)	8	44.1	22.3	18.8	6.4	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.1	8.1	7.3	7	3.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	8.8	7.4	6.5	2.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	7.9	7.3	7.1	3.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	8.7	7.4	6.6	2.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	7.6	7.2	7.2	3.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	8.6	7.4	6.7	3	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.2	7.7	7.3	7.2	3.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.4	8.6	7.3	6.6	3.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.3	2	2.1	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.4	2.1	2	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.3	2	2.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.4	2	2.1	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.5	2	2.1	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.4	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.4	1.9	2.1	0.9	
Scenario 6						
DLT rate	0.29	0.34	0.38	0.43	0.52	
Efficacy rate	0.3	0.32	0.35	0.38	0.4	
Utility 2	0	0	0	-0.1	-0.1	
Utility 3	46.4	45.6	45.8	45.6	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.4	16.4	1.9	1	0.1	15.2
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	67.1	15.7	1.4	0.7	0.1	15
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	66.5	14.6	2.5	0.8	0.1	15.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.1	14.3	1.4	0.6	0.1	15.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	68.3	13	2.3	0.9	0.2	15.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.3	12.3	1.5	0.7	0.1	15.1
U2 - Efficacy additive alpha prior Beta(1, 1)	68.3	13.2	2.4	0.9	0.2	15
U3 - Efficacy additive alpha prior Beta(1, 1)	70.7	11.7	1.6	0.6	0.1	15.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19	6.2	1.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.4	6.1	1.3	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.1	6	1.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	5.9	1.3	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.2	5.8	1.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.9	5.6	1.3	0.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	19.4	5.7	1.6	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.8	5.6	1.4	0.5	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	2.7	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	2.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	2.6	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	2.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.6	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	2.5	0.4	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	2.5	0.5	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	2.5	0.4	0.1	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 7						
DLT rate	0.16	0.2	0.26	0.31	0.36	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0	0.3	0.2	0.2	0.1	
Utility 3	45.6	62	59.6	57.6	55.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.2	55.8	22.4	6.8	0.5	1.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.3	59.6	19.6	4.1	0.3	1.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	55.8	22.5	6.3	0.7	1.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.6	60.8	18.3	4.1	0	1.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	54.1	21.7	7.5	0.7	1.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.6	56.6	18.7	4.7	0.1	1.3
U2 - Efficacy additive alpha prior Beta(1, 1)	14.5	54.9	21.7	7.1	0.7	1.1
U3 - Efficacy additive alpha prior Beta(1, 1)	18.2	57.7	18.3	4.6	0.1	1.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.6	14.5	5.1	1.4	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	15.1	4.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.8	14.2	5.2	1.5	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	15	4.3	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.9	13.6	5.3	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	14.3	4.6	1.3	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.9	13.7	5.2	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.5	14.5	4.4	1.3	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.9	1.3	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.9	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.9	1.3	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.9	1	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.9	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4	1	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.9	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.9	1	0.4	0	
Scenario 8						
DLT rate	0.32	0.37	0.42	0.48	0.52	
Efficacy rate	0.1	0.15	0.2	0.3	0.4	
Utility 2	-0.2	-0.2	-0.2	-0.2	-0.1	
Utility 3	33.2	34.2	35.2	38.8	43.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	44.6	9.5	0.9	0.1	0.1	44.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.6	8	0.9	0.1	0	44.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	45.1	8.7	1.1	0.1	0.1	44.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	46.6	7.1	1	0.1	0	45.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	45.1	8.4	1.4	0.1	0.1	44.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	46.4	8	0.6	0.1	0	44.9
U2 - Efficacy additive alpha prior Beta(1, 1)	45.6	7.9	1.5	0.1	0.1	44.8
U3 - Efficacy additive alpha prior Beta(1, 1)	46.9	7.7	0.7	0.1	0.1	44.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	5.4	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.3	5.3	1.2	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.1	5.4	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.4	5.2	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	5.3	1.2	0.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	5.2	1.1	0.4	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	17.3	5.3	1.2	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	17.5	5.1	1.1	0.4	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.7	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	2.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.7	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	2.6	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	2.7	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	2.7	0.4	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.7	0.4	0.2	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	2.6	0.4	0.2	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 9						
DLT rate	0.15	0.18	0.22	0.26	0.29	
Efficacy rate	0.05	0.08	0.2	0.25	0.5	
Utility 2	-0.1	-0.1	0	0	0.2	
Utility 3	37	37.6	43.2	44.6	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.9	20.1	23.3	14.5	16.2	7
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.9	22.3	18.6	12.8	13.6	6.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.7	21.3	21.9	15.1	16	7
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.8	21.8	18.5	14.2	12.6	7.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.7	20.6	22.7	14	16	7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.7	21	19.8	13	13.5	7
U2 - Efficacy additive alpha prior Beta(1, 1)	18	21.5	23.7	14.1	15.7	7
U3 - Efficacy additive alpha prior Beta(1, 1)	25.5	21.2	19.4	12.7	13.8	7.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	8.3	7	5.9	4.9	3.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9	6.8	5.8	4.8	3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.5	6.9	6	4.8	3.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.2	6.7	5.7	4.7	3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.5	6.8	5.9	4.8	3.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.2	6.7	5.7	4.7	3.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	8.6	6.8	5.9	4.8	3.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.2	6.7	5.7	4.6	3.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4	1.7	1.6	0.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.6	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	1.7	1.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.9	1.6	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.6	1.7	0.7	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	1.6	1.6	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.7	1.7	0.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	1.6	1.6	0.7	
Scenario 10						
DLT rate	0.23	0.28	0.32	0.39	0.43	
Efficacy rate	0.05	0.4	0.42	0.45	0.5	
Utility 2	-0.2	0.1	0.1	0.1	0.1	
Utility 3	33.8	52.8	52.4	51.4	52.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.4	38.3	12.4	2.9	0.4	27.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.4	38.3	11.3	2.3	0.3	28.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.4	37.6	12.5	2.5	0.4	28.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19	39.1	11	2.4	0.4	28.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.3	39.3	12.2	2.7	0.5	28
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.2	39.1	10.5	2.1	0.3	27.8
U2 - Efficacy additive alpha prior Beta(1, 1)	17.8	39.2	12.2	2.7	0.4	27.7
U3 - Efficacy additive alpha prior Beta(1, 1)	19.7	38.2	11.3	2.1	0.3	28.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	10.2	3.8	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.9	10.4	3.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	10.2	3.8	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12	10.4	3.5	1.1	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.8	10.1	3.8	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.1	10.3	3.5	1.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	10.1	3.8	1.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.1	10.4	3.6	1.1	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.3	1	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4.2	0.9	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.3	1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.3	0.9	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.3	1	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.3	0.9	0.4	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.3	1	0.4	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.3	0.9	0.4	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 11						
DLT rate	0.22	0.32	0.44	0.59	0.67	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.4	-0.5	
Utility 3	43.2	39.2	34.4	28.4	25.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	68.4	21.5	2.5	0.1	0	7.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	70.8	20	2	0.1	0	7.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	69.1	21.4	2.2	0	0	7.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	70.7	20.1	1.8	0	0	7.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	70	19.8	3	0	0	7.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	73.6	17.3	1.6	0	0	7.5
U2 - Efficacy additive alpha prior Beta(1, 1)	70	20.6	2.3	0.1	0	7
U3 - Efficacy additive alpha prior Beta(1, 1)	72.9	17.8	1.8	0	0	7.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.1	7.7	2.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	7.5	2.1	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.3	7.7	2.3	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.9	7.3	2.1	0.7	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	7.4	2.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19	7.2	2.1	0.7	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.5	7.4	2.4	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	19	7.2	2.1	0.7	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	3.6	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.7	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	3.6	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	3.6	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	3.5	0.7	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	3.6	0.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	3.5	0.7	0.3	0	
Scenario 12						
DLT rate	0.06	0.09	0.18	0.29	0.44	
Efficacy rate	0.05	0.1	0.12	0.4	0.6	
Utility 2	0	0	-0.1	0.1	0.2	
Utility 3	40.6	42.4	40	52.4	58.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.7	16.3	11.4	44.7	15.6	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.9	16.2	10.5	41.7	14.4	0.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	15.9	11.2	45.2	16	0.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.2	16.5	9.7	40.7	14.6	0.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.4	15.4	12.7	43.7	15.5	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.4	17.3	9.5	41	14.5	0.3
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	16.2	12.6	43.8	15.9	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	17.9	15.3	9.7	42.1	14.6	0.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.8	6.1	9.1	4.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.4	5.9	5.9	9	3.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5	5.8	6	9	4.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.5	5.9	5.9	8.9	3.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	5.8	6	8.8	4.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.5	5.9	5.8	8.8	3.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.1	5.8	5.9	8.9	4.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.5	5.9	5.8	8.8	3.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	3.3	1.7	2.4	1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	3.3	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3.4	1.7	2.4	1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	3.4	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.4	1.7	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	3.4	1.6	2.3	0.9	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.4	1.7	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.5	3.4	1.6	2.3	0.9	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 13						
DLT rate	0.21	0.29	0.3	0.39	0.46	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.2	0	0	0	-0.1	
Utility 3	34.6	46.4	47.2	45.4	45.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	20.3	40.3	11.4	3.5	0.5	24
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.9	39.7	10.6	2.5	0.4	24.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.7	39.6	11	3.9	0.5	24.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.7	41	8.5	2.9	0.4	24.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.6	39.2	11.6	3.5	0.5	24.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23	40.5	9.2	2.2	0.5	24.6
U2 - Efficacy additive alpha prior Beta(1, 1)	21.3	38.9	11.5	3	0.5	24.8
U3 - Efficacy additive alpha prior Beta(1, 1)	24.1	39.6	8.6	2.4	0.5	24.8
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	12	9.6	4.1	1.7	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	9.8	3.8	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.9	9.7	4	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.2	9.8	3.8	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.9	9.6	4.1	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.3	9.7	3.8	1.6	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.9	9.6	4	1.8	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.3	9.7	3.8	1.6	0.3	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.5	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.5	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.5	1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.5	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4.5	1.1	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.5	1	0.6	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	4.5	1	0.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.5	1	0.6	0.1	
Scenario 14						
DLT rate	0.09	0.15	0.23	0.28	0.39	
Efficacy rate	0.05	0.08	0.12	0.3	0.4	
Utility 2	0	-0.1	-0.1	0	0	
Utility 3	39.4	38.8	38	46.8	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.6	21.2	19.6	28.5	9.6	1.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.6	24.1	14.1	25.4	7.5	1.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.6	21.4	18.8	29	9.9	1.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.9	23.9	14.4	26.1	7.4	1.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	20	19.5	29.4	9.7	1.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.8	22.8	14.9	25.8	8	1.7
U2 - Efficacy additive alpha prior Beta(1, 1)	18.7	21.4	19.2	28.7	10.4	1.6
U3 - Efficacy additive alpha prior Beta(1, 1)	26.5	23.5	15.3	25.3	7.8	1.6
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.7	7	6.3	6.8	3.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.3	7	5.9	6.7	2.9	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.8	6.9	6.2	6.8	3.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.5	6.8	5.9	6.7	2.9	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.8	6.9	6.2	6.8	3.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.6	6.8	5.9	6.7	2.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	6.9	6.9	6.1	6.7	3.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	7.7	6.8	5.8	6.6	2.9	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.8	1.8	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	1.7	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.8	1.8	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.8	1.7	2	0.7	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.8	1.8	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.9	1.7	2	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.8	1.7	2.1	0.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.9	1.7	2	0.7	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 15						
DLT rate	0.13	0.2	0.32	0.39	0.46	
Efficacy rate	0.1	0.1	0.1	0.1	0.1	
Utility 2	0	-0.1	-0.2	-0.3	-0.4	
Utility 3	40.8	38	33.2	30.4	27.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	46.8	35.2	12.3	3	0.1	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	54.3	32.3	8.5	2.2	0.1	2.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	49.5	33.9	10.9	3.1	0.2	2.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	57	29.6	7.8	2.6	0.1	2.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	51.2	32.9	10.1	3.1	0.2	2.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	58.7	28.5	7.7	2.2	0.1	2.8
U2 - Efficacy additive alpha prior Beta(1, 1)	49.6	33.1	11.1	3.4	0.3	2.5
U3 - Efficacy additive alpha prior Beta(1, 1)	58.6	27.7	8.2	2.4	0.1	3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.2	9.2	5.4	3.1	0.8	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.1	8.8	5.1	3	0.8	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.4	9.1	5.4	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	12.3	8.7	5	3	0.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11.6	9	5.3	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	12.5	8.6	5	2.9	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	11.5	9.1	5.3	3.1	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	12.5	8.6	5	3	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.4	1.8	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.2	4.2	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.2	4.4	1.8	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.3	4.3	1.7	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.2	4.4	1.7	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.3	4.3	1.6	1.2	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	1.2	4.5	1.7	1.3	0.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.3	4.3	1.6	1.2	0.3	
Scenario 16						
DLT rate	0.02	0.06	0.09	0.14	0.26	
Efficacy rate	0.2	0.2	0.2	0.2	0.2	
Utility 2	0.2	0.1	0.1	0.1	-0.1	
Utility 3	51.2	49.6	48.4	46.4	41.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	36	27.9	14.9	13.9	7.3	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	41.2	30.4	14.1	10	4.3	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	35.3	27.2	16.6	14.1	6.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	40.7	29.9	14.8	10.4	4.2	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	36	27.7	15	12.8	8.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	42.8	29.3	12.5	11	4.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	36.4	26.4	15.7	13	8.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	41.6	30.4	13.4	10.2	4.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.3	5.3	6.1	8.2	6.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.7	6.2	8	5.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.4	5.1	6.1	8.1	6.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	5.6	6.2	7.9	5.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.4	4.9	6.1	8.1	6.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.9	5.2	6.3	8	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.4	4.9	6.1	8	6.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	5	5.3	6.2	7.9	5.6	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.2	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.9	2.2	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2.1	3	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	2.9	2.2	2.8	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	2.1	3	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	2.2	2.9	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	2.1	2.9	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	2.9	2.1	2.9	1.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 17						
DLT rate	0.11	0.23	0.39	0.58	0.76	
Efficacy rate	0.3	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.3	0.1	-0.1	-0.3	
Utility 3	53.6	60.8	54.4	46.8	39.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.7	66.5	14	0.7	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.6	67.9	10.1	0.3	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.4	65.4	12.3	0.8	0	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	24.6	66	9	0.3	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.6	61.9	13.5	0.9	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	25.9	63.4	9.9	0.7	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	23.2	62.7	13.2	0.8	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	26.4	63.7	9.1	0.7	0	0.1
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	15.9	4.1	0.4	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.3	16.2	3.2	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	15.4	4.1	0.5	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	15.7	3.2	0.4	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10	14.8	4.5	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	15.1	3.6	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.1	14.7	4.4	0.7	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.9	15.2	3.5	0.4	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4.1	1.2	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4	0.8	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	1.1	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4.1	0.8	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	0.9	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.9	1.2	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	0.9	0.2	0	
Scenario 18						
DLT rate	0.02	0.04	0.1	0.2	0.35	
Efficacy rate	0.05	0.1	0.15	0.2	0.5	
Utility 2	0	0.1	0.1	0	0.1	
Utility 3	42.2	44.4	45	44	56	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	11.6	10.6	18.4	49.7	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.1	15.2	9.5	16.1	43.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	8.6	11.8	10.5	18.5	50.6	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.8	14.2	10.1	15.2	44.7	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	8.7	11	10.4	19.6	50.3	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.6	14.4	9.7	15.2	45.1	0
U2 - Efficacy additive alpha prior Beta(1, 1)	8.3	11.4	10.6	18.2	51.5	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15	14.6	10.1	14.8	45.5	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	4.1	5.2	7.3	9.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.2	5.4	7.5	9.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	4.1	5.2	7.1	10.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.3	5.4	7.3	9.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	4.1	5.2	7.1	10.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.7	4.3	5.4	7.1	9.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.5	4.1	5.2	7.1	10.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.7	4.3	5.3	7.1	9.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.5	1.7	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.8	1.6	2.5	1.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.8	1.6	2.5	1.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.6	2.5	1.6	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 19						
DLT rate	0.11	0.26	0.39	0.6	0.77	
Efficacy rate	0.08	0.3	0.32	0.35	0.4	
Utility 2	0	0	-0.1	-0.2	-0.4	
Utility 3	40.4	47.6	43.6	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.2	60	16.7	0.7	0	4.4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.2	58.8	14.1	0.5	0	4.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.2	60.6	15.4	0.7	0	4.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	21.8	60.4	13.1	0.4	0	4.3
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.3	61.6	13.9	0.7	0	4.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	21.5	60	13.9	0.7	0	3.9
U2 - Efficacy additive alpha prior Beta(1, 1)	19.8	60.8	14.4	0.6	0	4.4
U3 - Efficacy additive alpha prior Beta(1, 1)	21.9	60.1	13	1	0	4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.3	13.2	5.6	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.6	13.4	5.3	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	13.3	5.6	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.7	13.3	5.2	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.4	13.1	5.5	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.7	13.2	5.3	1.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.4	13	5.6	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	9.7	13.2	5.2	1.4	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	4.9	1.7	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.9	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	5	1.6	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.9	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5	1.5	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5	1.5	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	5	1.5	0.7	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	5	1.5	0.6	0	
Scenario 20						
DLT rate	0.07	0.14	0.31	0.41	0.52	
Efficacy rate	0.05	0.25	0.25	0.25	0.25	
Utility 2	0	0.1	-0.1	-0.2	-0.3	
Utility 3	40.2	49.4	42.6	38.6	34.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.2	50.5	30.2	8.9	0.7	0.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.7	53.8	24.1	7.5	0.4	0.5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.8	50.2	30.2	8.4	0.8	0.6
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.5	53	24.9	7.5	0.6	0.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	11	50	29.4	8.3	0.8	0.5
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.6	52.7	24	7.5	0.6	0.6
U2 - Efficacy additive alpha prior Beta(1, 1)	11.3	50.6	28.8	8.1	0.6	0.6
U3 - Efficacy additive alpha prior Beta(1, 1)	14.4	52.6	24.4	7.3	0.8	0.5
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.2	11	8.4	4.2	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	11.4	7.9	4	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.3	10.9	8.4	4.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	11.4	7.9	3.9	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.4	10.7	8.2	4.4	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.8	11.2	7.9	4	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.4	10.6	8.2	4.4	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.8	11.2	7.8	4.1	1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.4	4.1	2.4	1.5	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.5	4	2.3	1.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.5	4.1	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	4.1	2.2	1.5	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.5	4.1	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4.2	2.2	1.5	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.5	4.1	2.3	1.6	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	4.2	2.2	1.5	0.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 21						
DLT rate	0.37	0.56	0.69	0.83	0.88	
Efficacy rate	0.5	0.5	0.5	0.5	0.5	
Utility 2	0.1	-0.1	-0.2	-0.3	-0.4	
Utility 3	55.2	47.6	42.4	36.8	34.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	54	1.2	0	0	0	44.8
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	54.4	0.8	0	0	0	44.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	54.3	0.8	0	0	0	44.9
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	54.7	0.8	0	0	0	44.5
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	54.7	0.9	0	0	0	44.4
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	54.3	0.8	0	0	0	44.9
U2 - Efficacy additive alpha prior Beta(1, 1)	54.8	0.8	0	0	0	44.4
U3 - Efficacy additive alpha prior Beta(1, 1)	54.9	0.7	0	0	0	44.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.9	2.3	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	20	2.2	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	20	2.2	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	20.1	2.2	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	20	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	20.1	2.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	20	2.2	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	20.1	2.1	0	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	1.1	0	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	1.1	0	0	0	
Scenario 22						
DLT rate	0.07	0.27	0.5	0.56	0.78	
Efficacy rate	0.08	0.25	0.25	0.5	0.5	
Utility 2	0	0	-0.2	-0.1	-0.3	
Utility 3	42	44.2	35	47.6	38.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	21.6	64.9	10	1.4	0	2.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.3	61.1	9.6	1.7	0	2.3
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.6	62.9	10.6	1.8	0	2.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	60.3	8.9	2	0	2.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.9	63.6	9.8	1.5	0	2.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	26.5	59.9	9.8	1.7	0	2.1
U2 - Efficacy additive alpha prior Beta(1, 1)	22.4	63.7	10	1.8	0	2.1
U3 - Efficacy additive alpha prior Beta(1, 1)	27.2	59.1	10	1.8	0	1.9
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.4	13.9	5.1	1.5	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.8	13.8	4.8	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	13.7	5.1	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	13.6	4.8	1.4	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	13.6	5.1	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.1	13.4	4.8	1.5	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	13.5	5.1	1.6	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.1	13.4	4.9	1.5	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.1	4.9	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1.1	5	1.6	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	5.1	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	5.1	1.6	0.5	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	5.1	1.6	0.6	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	5.1	1.6	0.6	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 23						
DLT rate	0.32	0.55	0.8	0.94	0.97	
Efficacy rate	0.05	0.05	0.05	0.05	0.05	
Utility 2	-0.3	-0.5	-0.8	-0.9	-0.9	
Utility 3	30.2	21	11	5.4	4.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.5	2	0	0	0	80.5
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.7	1.7	0	0	0	80.6
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.8	2	0	0	0	80.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	2	0	0	0	80.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.2	2.1	0	0	0	80.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	17.8	1.5	0	0	0	80.7
U2 - Efficacy additive alpha prior Beta(1, 1)	17.5	2.4	0	0	0	80.1
U3 - Efficacy additive alpha prior Beta(1, 1)	17.8	1.9	0	0	0	80.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.4	4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.5	3.9	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.4	4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	15.5	3.9	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.4	4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	15.5	3.9	0.3	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	15.5	4	0.3	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	15.5	3.9	0.3	0	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	2.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	2.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	2.3	0.1	0	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.4	0.1	0	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.6	2.3	0.1	0	0	
Scenario 24						
DLT rate	0.01	0.02	0.05	0.1	0.25	
Efficacy rate	0.05	0.08	0.2	0.5	0.5	
Utility 2	0	0.1	0.1	0.4	0.2	
Utility 3	42.6	44	50	66	60	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1.3	0.8	3	53.4	41.5	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.3	2	3.7	57.2	34.8	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	0.7	3.4	54.8	40.2	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	2	1.7	3.8	58.1	34.4	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	0.7	3.9	52.9	41.4	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.6	1.7	5.3	56.8	34.6	0
U2 - Efficacy additive alpha prior Beta(1, 1)	1.1	0.9	3.7	54.2	40.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	1.8	1.7	4.5	57.6	34.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	3.8	12.7	7.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.1	3.2	4	13.4	6.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	3.8	12.8	7.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.1	3.2	4	13.3	6.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.1	3.9	12.3	7.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.1	3.2	4.1	12.9	6.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.1	3.9	12.4	7.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.1	3.2	4.1	13	6.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0	2.4	1.1	2.4	1.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.4	1.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0	2.4	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.4	1	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0	2.5	1.1	2.5	1.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1	2.5	1.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0	2.4	1.1	2.5	1.3	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 25						
DLT rate	0.09	0.18	0.37	0.63	0.77	
Efficacy rate	0.2	0.25	0.4	0.45	0.5	
Utility 2	0.1	0.1	0	-0.2	-0.3	
Utility 3	48.4	47.8	49.2	41.8	39.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	26.6	37.4	32.6	3.3	0	0.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	33.1	35.6	28.5	2.7	0	0.1
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	28.5	36.6	31.7	3.1	0	0.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	33.4	36.7	27.4	2.4	0	0.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	30	35.3	31.7	2.9	0	0.1
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	35.1	35	27.2	2.6	0	0.1
U2 - Efficacy additive alpha prior Beta(1, 1)	30.9	34.9	31	3.1	0	0.1
U3 - Efficacy additive alpha prior Beta(1, 1)	34.6	35.7	26.7	2.7	0.1	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.1	10.5	8.2	2	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.7	10.6	7.8	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.4	10.3	8.1	2	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10	10.4	7.6	1.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.5	10	8.2	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	10.2	7.7	1.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.5	10.1	8.1	2.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.3	10.2	7.5	1.9	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4	2.1	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4	1.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.9	2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4	1.9	0.8	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4	2	0.9	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4	1.8	0.8	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4	2	0.8	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4	1.8	0.8	0	
Scenario 26						
DLT rate	0.15	0.22	0.3	0.62	0.8	
Efficacy rate	0.1	0.2	0.35	0.45	0.55	
Utility 2	-0.1	0	0.1	-0.2	-0.2	
Utility 3	40	43.2	49	42.2	41	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	25.2	40.6	27.3	3	0	3.9
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	30.4	39.2	23.9	2.6	0	3.9
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	25.6	40.1	27.9	2.9	0	3.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	30.3	39.7	23.6	2.2	0	4.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.1	39	26.5	2.5	0	3.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	33.6	36.1	23.8	2.7	0	3.8
U2 - Efficacy additive alpha prior Beta(1, 1)	27.5	38.4	26.9	3.1	0	4.1
U3 - Efficacy additive alpha prior Beta(1, 1)	32.7	36.9	23.7	2.5	0	4.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	10.1	7.1	2.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.6	10	6.8	2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.1	10	7.1	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.7	9.9	6.8	2.1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	9.9	7	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	9.8	6.8	2.1	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.3	9.7	7.1	2.3	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	9.8	6.8	2.1	0.1	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.4	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	4.3	1.6	0.9	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.4	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	4.4	1.6	0.9	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.4	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	4.4	1.6	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	4.4	1.6	1	0.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.4	1.5	0.9	0.1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 27						
DLT rate	0.11	0.17	0.3	0.5	0.62	
Efficacy rate	0.1	0.2	0.4	0.42	0.45	
Utility 2	0	0	0.1	-0.1	-0.2	
Utility 3	41.6	45.2	52	45.2	42.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.1	30.6	43.4	7.6	0	1.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	21	32.5	40	5.2	0.1	1.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.7	31.4	42.7	6.9	0	1.3
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.3	31.8	39.6	5.1	0.1	1.1
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.4	32.2	42.5	5.5	0.1	1.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	22.7	30.4	39.7	5.8	0.1	1.3
U2 - Efficacy additive alpha prior Beta(1, 1)	18.4	31.9	42.6	5.8	0	1.3
U3 - Efficacy additive alpha prior Beta(1, 1)	22.8	29.9	41.2	4.8	0	1.3
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.3	9.3	9.7	3.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	9.3	9.4	3.1	0.3	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.4	9.2	9.6	3.3	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	7.8	9.3	9.5	3	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	7.5	8.9	9.8	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	8	9.1	9.4	3	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	7.5	9	9.7	3.2	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	8	9	9.5	3	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	4	2.1	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.7	3.9	2.1	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.9	2.1	1.3	0.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	4	2.1	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	4	2.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	4	2	1.2	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	4	2.1	1.2	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	4	2	1.2	0.1	
Scenario 28						
DLT rate	0.2	0.3	0.5	0.6	0.77	
Efficacy rate	0.05	0.3	0.32	0.35	0.4	
Utility 2	-0.1	0	-0.2	-0.2	-0.4	
Utility 3	35	46	39.2	37	33.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.4	43.1	4.3	0.1	0	25.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	28.4	42.5	3.3	0	0	25.8
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.4	43.5	3.2	0.1	0	25.8
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.3	43	3.5	0	0	26.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	27.7	43.9	3.1	0.1	0	25.2
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.1	42.2	3.3	0	0	26.4
U2 - Efficacy additive alpha prior Beta(1, 1)	27.2	43.8	3.7	0.1	0	25.2
U3 - Efficacy additive alpha prior Beta(1, 1)	28.5	42.1	3.3	0.1	0	26
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.1	11.2	2.8	0.6	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	13.4	11	2.7	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.2	11.1	2.8	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	13.4	11.1	2.7	0.6	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	11	2.8	0.7	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.5	10.9	2.7	0.6	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	13.2	10.9	2.9	0.7	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	13.5	11	2.6	0.6	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.8	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	4.6	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.7	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	4.7	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	4.8	0.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	4.7	0.9	0.3	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.8	1	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	4.7	0.9	0.3	0	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 29						
DLT rate	0.19	0.3	0.56	0.74	0.86	
Efficacy rate	0.4	0.4	0.4	0.4	0.4	
Utility 2	0.2	0.1	-0.2	-0.3	-0.5	
Utility 3	56.4	52	41.6	34.4	29.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	62.7	32	2.6	0.1	0	2.6
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	65.7	30.1	1.8	0	0	2.4
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	63.2	31.4	2.8	0.1	0	2.5
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	68.3	27.3	2	0	0	2.4
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	67.7	27.6	1.9	0.1	0	2.7
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	70.6	25.1	1.8	0.1	0	2.4
U2 - Efficacy additive alpha prior Beta(1, 1)	67.6	27.6	2.2	0.2	0	2.4
U3 - Efficacy additive alpha prior Beta(1, 1)	70.1	25.4	2	0.1	0	2.4
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	18.3	9.2	1.7	0.2	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	19.1	9	1.3	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.4	9	1.8	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	19.7	8.4	1.2	0.2	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	18.8	8.4	1.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.9	8.1	1.3	0.2	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	18.9	8.4	1.9	0.3	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	19.8	8.1	1.3	0.2	0	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.3	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	3.1	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	3.3	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	3.1	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	3.1	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	3	0.5	0.1	0	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	3.1	0.8	0.1	0	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	3	0.5	0.1	0	
Scenario 30						
DLT rate	0.05	0.08	0.17	0.27	0.54	
Efficacy rate	0.12	0.15	0.2	0.4	0.5	
Utility 2	0.1	0.1	0	0.1	0	
Utility 3	45.2	45.8	45.2	53.2	48.4	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	15.9	14.2	15.5	42.8	11.6	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	22.8	16.8	14	37.4	9	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	17.6	12.1	15.6	43.9	10.8	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	22.6	17.3	13.3	38.2	8.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	16.1	12.8	17.3	44.1	9.7	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.2	17	13.2	38.1	8.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	15.9	12.9	17	43.8	10.4	0
U3 - Efficacy additive alpha prior Beta(1, 1)	23	17.1	13.7	38.3	7.9	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	5.5	6.6	9.9	3.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.9	5.6	6.6	9.8	3.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.7	5.3	6.6	9.9	3.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.1	5.5	6.5	9.8	3.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	5.2	6.6	9.8	3.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.1	5.4	6.5	9.7	3.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	5.2	6.5	9.9	3.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	5.5	6.5	9.7	3.2	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	2.9	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3	2	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.4	3	1.9	2.3	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	2.9	1.9	2.4	1.1	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.4	3.1	1.9	2.3	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.3	3	1.9	2.3	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.4	3.1	1.9	2.3	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 31						
DLT rate	0.04	0.06	0.1	0.2	0.6	
Efficacy rate	0.2	0.5	0.5	0.5	0.5	
Utility 2	0.2	0.4	0.4	0.3	-0.1	
Utility 3	50.4	67.6	66	62	46	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	2.8	40.2	29.2	24.5	3.3	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	46.9	28.1	19.6	1.5	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.3	39.8	28.5	24.4	4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	46.4	27.3	18.8	2.6	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.5	40.2	27.6	23.2	4.5	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.3	46.2	25.8	19.3	2.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	4	40.7	28.5	22.8	4	0
U3 - Efficacy additive alpha prior Beta(1, 1)	6	44.4	27.5	19.5	2.6	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.9	13.3	7.7	4.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4	14.2	7.5	3.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.1	12.9	7.5	4.7	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.3	13.9	7.1	4.1	0.6	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4	12.2	7.7	4.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.2	13.2	7.4	4.5	0.8	
U2 - Efficacy additive alpha prior Beta(1, 1)	4	12.2	7.8	4.8	1.1	
U3 - Efficacy additive alpha prior Beta(1, 1)	4.2	13.1	7.4	4.4	0.8	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.8	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.9	1.6	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.8	1.7	1.1	0.3	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.9	1.5	0.9	0.3	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.9	1.6	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.9	1.5	1	0.3	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.9	1.6	1.1	0.4	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	1.5	1	0.3	
Scenario 32						
DLT rate	0.02	0.05	0.11	0.3	0.36	
Efficacy rate	0.08	0.15	0.3	0.3	0.3	
Utility 2	0.1	0.1	0.2	0	-0.1	
Utility 3	44	47	53.6	46	43.6	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	11.4	15.4	31.2	32.2	9.8	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	16.4	19.1	32	26.5	6	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	11	13.8	35.7	30.2	9.3	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	16.4	18.3	32.6	26.4	6.3	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.5	13.6	36.4	28.9	10.6	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	14.7	19.4	33.2	26.2	6.5	0
U2 - Efficacy additive alpha prior Beta(1, 1)	11.2	13.4	36.2	29.4	9.8	0
U3 - Efficacy additive alpha prior Beta(1, 1)	15	18.1	34.6	25.6	6.7	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.5	4.5	8.6	9	4.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.6	4.7	9.1	8.8	3.7	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.6	4.3	8.6	8.9	4.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.7	4.6	9.1	8.7	3.8	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.6	4.3	8.7	8.6	4.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.8	4.6	9	8.6	4	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.6	4.3	8.7	8.6	4.7	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.8	4.6	9.2	8.4	4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	2.8	2.1	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	2.8	2	2.8	1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.8	1.9	2.9	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	2.9	1.9	2.8	1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.8	1.9	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.2	2.9	2	2.8	1	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 33						
DLT rate	0.01	0.02	0.09	0.32	0.38	
Efficacy rate	0.05	0.08	0.2	0.4	0.5	
Utility 2	0	0.1	0.1	0.1	0.1	
Utility 3	42.6	44	48.4	51.2	54.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.8	6.8	11.4	48.8	26.2	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	12.2	7.3	12.2	46.2	22.1	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.4	6.8	11.4	48.7	26.7	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.4	8.1	12.4	46.6	22.5	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.8	6.9	11.4	46.8	28.1	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.6	8.8	12.3	46.3	22	0
U2 - Efficacy additive alpha prior Beta(1, 1)	6.2	6.3	11.6	48.5	27.4	0
U3 - Efficacy additive alpha prior Beta(1, 1)	11.5	9.1	12.1	44.9	22.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.3	3.8	6.1	10.8	5.9	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	3.4	4	6.3	11	5.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.4	3.8	6.1	10.7	6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	3.5	3.9	6.2	10.9	5.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.4	3.8	6.1	10.3	6.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	3.5	3.9	6.2	10.8	5.6	
U2 - Efficacy additive alpha prior Beta(1, 1)	3.4	3.8	6	10.5	6.3	
U3 - Efficacy additive alpha prior Beta(1, 1)	3.5	3.9	6.3	10.6	5.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.1	2.7	1.6	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.1	2.7	1.6	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.1	2.7	1.6	2.8	1.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.5	2.8	1.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.1	2.7	1.6	2.8	1	
Scenario 34						
DLT rate	0.05	0.13	0.3	0.36	0.42	
Efficacy rate	0.05	0.1	0.3	0.32	0.35	
Utility 2	0	0	0	0	-0.1	
Utility 3	41	40.8	46	44.8	44.2	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	13	17.8	49.2	17.1	2.6	0.3
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	17.2	19.8	45.3	15.8	1.7	0.2
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	14.4	18.2	47.2	17.3	2.7	0.2
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	18.8	18.3	43.7	16.9	2.1	0.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.2	18.4	48.3	16.7	3.1	0.3
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	19.8	17.4	43.9	17	1.7	0.2
U2 - Efficacy additive alpha prior Beta(1, 1)	14.3	16.7	50.2	15.4	3.1	0.3
U3 - Efficacy additive alpha prior Beta(1, 1)	20.8	17.2	43.7	16.2	1.9	0.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	5.6	7.1	10	5.7	1.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	6.1	7.1	9.9	5.5	1.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	5.7	7.1	9.9	5.5	1.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	6.3	7.1	9.8	5.4	1.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	5.7	7.1	9.8	5.6	1.8	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	6.4	7	9.9	5.4	1.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	5.8	7	9.8	5.5	1.8	
U3 - Efficacy additive alpha prior Beta(1, 1)	6.4	6.9	9.8	5.4	1.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.7	2.3	1.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.6	3.6	2.2	1.8	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.6	3.7	2.2	1.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.7	3.7	2.1	1.8	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.6	3.7	2.2	1.9	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.7	3.7	2.1	1.8	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.6	3.7	2.2	1.9	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.7	3.7	2.1	1.8	0.4	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 35						
DLT rate	0.03	0.09	0.3	0.4	0.45	
Efficacy rate	0.2	0.45	0.45	0.45	0.45	
Utility 2	0.2	0.4	0.1	0.1	0	
Utility 3	50.8	63.4	55	51	49	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	7.8	44.5	35.2	11.1	1.4	0
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	48.4	32	8.2	0.7	0
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.3	42.9	35.3	10.1	2.4	0
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	11.6	48.4	31	7.8	1.2	0
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.3	44.9	32.9	10.7	2.2	0
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	13.5	47.1	29.8	8.2	1.4	0
U2 - Efficacy additive alpha prior Beta(1, 1)	9	43.9	32.9	12.1	2.1	0
U3 - Efficacy additive alpha prior Beta(1, 1)	13.5	47.5	29.2	8.4	1.4	0
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.4	12.9	8.9	3.2	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	4.6	13.7	8.3	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.6	12.6	8.7	3.4	0.8	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	4.9	13.5	8.1	2.9	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	4.7	12.1	8.8	3.6	0.9	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	5	13	8.3	3	0.7	
U2 - Efficacy additive alpha prior Beta(1, 1)	4.7	12.1	8.7	3.6	0.9	
U3 - Efficacy additive alpha prior Beta(1, 1)	5.1	13	8.2	3	0.7	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.2	3.2	2.1	1	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.3	3.1	2	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.2	3.2	2.1	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.3	3.2	1.9	1	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.2	3.3	1.9	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.3	3.3	1.9	0.9	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.2	3.3	2	1.1	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.3	3.2	1.9	0.9	0.2	
Scenario 36						
DLT rate	0.13	0.29	0.35	0.4	0.43	
Efficacy rate	0.05	0.2	0.22	0.3	0.35	
Utility 2	-0.1	-0.1	-0.1	-0.1	-0.1	
Utility 3	37.8	40.4	39.2	42	43.8	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	24	48.5	12.7	3.9	0.8	10.1
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.3	48	10.7	3.6	0.4	10
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.8	49	12.6	3.7	0.8	10.1
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	26.7	48.3	11.6	2.8	0.4	10.2
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	23.7	49.8	11.8	3.7	1.1	9.9
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.1	45.9	10.4	3.7	0.6	11.3
U2 - Efficacy additive alpha prior Beta(1, 1)	24.4	48.9	12.4	3.7	0.8	9.8
U3 - Efficacy additive alpha prior Beta(1, 1)	28.8	46.5	10.2	3.6	0.7	10.2
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.2	11.1	4.9	2.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.7	11	4.6	2.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	11	5	2.5	0.6	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.8	10.9	4.7	2.4	0.5	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.3	10.9	4.9	2.4	0.6	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.8	10.8	4.6	2.4	0.5	
U2 - Efficacy additive alpha prior Beta(1, 1)	10.4	10.9	4.9	2.5	0.6	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.8	10.8	4.6	2.4	0.5	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.8	5.1	1.4	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	0.9	5	1.3	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.8	5.1	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	5	1.4	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.8	5.2	1.4	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	0.9	5.1	1.4	0.9	0.2	
U2 - Efficacy additive alpha prior Beta(1, 1)	0.8	5.2	1.4	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	0.9	5.1	1.4	0.9	0.2	

Phase I/II Operating Characteristics

Efficacy additive alpha prior comparison - allocation = one_stage - IPDE alpha = 0.6 - both utilities - N = 30

	Dose 1	Dose 2	Dose 3	Dose 4	Dose 5	Stop %
Scenario 37						
DLT rate	0.08	0.3	0.4	0.45	0.5	
Efficacy rate	0.05	0.2	0.2	0.2	0.2	
Utility 2	0	-0.1	-0.2	-0.2	-0.3	
Utility 3	39.8	40	36	34	32	
OBD selection (%)						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	23.4	58.5	12.2	1.9	0	4
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	27.4	55.5	10.9	1.1	0.1	5
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	23.4	58.8	11.9	1.5	0	4.4
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	27.8	55.8	10	1.5	0	4.9
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	25	56.3	12	2.1	0	4.6
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	28.5	54.5	10.3	1.8	0	4.9
U2 - Efficacy additive alpha prior Beta(1, 1)	24.3	57.7	11.6	2	0.1	4.3
U3 - Efficacy additive alpha prior Beta(1, 1)	27.6	55.1	10.9	1.7	0	4.7
Mean patients treated						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	9.5	12.5	5.1	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	10.1	12.3	4.9	2.1	0.4	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	9.5	12.3	5.2	2.2	0.4	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	10.2	12.2	4.9	2	0.4	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	9.6	12.2	5.2	2.3	0.5	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	10.2	12	4.9	2.1	0.4	
U2 - Efficacy additive alpha prior Beta(1, 1)	9.6	12.2	5.2	2.2	0.5	
U3 - Efficacy additive alpha prior Beta(1, 1)	10.2	12.1	4.9	2	0.4	
Mean IPDE patients						
U2 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5.4	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.15, 0.85)	1	5.2	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.3, 0.7)	0.9	5.4	1.6	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.3, 0.7)	1	5.3	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(0.5, 0.5)	1	5.4	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(0.5, 0.5)	1.1	5.3	1.5	0.8	0.1	
U2 - Efficacy additive alpha prior Beta(1, 1)	1	5.5	1.5	0.9	0.2	
U3 - Efficacy additive alpha prior Beta(1, 1)	1.1	5.4	1.5	0.8	0.1	